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KIRIBATI

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STAFF REPORT FOR THE 2018 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)[1]

The 2018 Debt Sustainability Analysis (DSA) indicates that Kiribati remains at high risk of debt distress, although the debt burden indicators improved somewhat thanks to the improved fiscal position and associated increase in fiscal buffers over the last few years.[2] Further progress in structural and fiscal reforms is needed to improve debt trajectories. The ratios of the present value (PV) of external public and publicly-guaranteed (PPG) debt to GDP and to exports are currently below their respective policy dependent indicative thresholds. However, these ratios are increasing and expected to exceed their indicative thresholds in the medium to the long term. Stress tests confirm the vulnerability of the debt position to plausible shocks. Although Kiribati does not currently face debt servicing risks, helped by historically high revenue from fishing license fees and large cash buffers, risks from a possible reversal of favorable weather or global market conditions and from contingent liabilities call for continued fiscal prudence. Containing the risk of debt distress requires continuation of grants to support the country's large development needs, and implementation of fiscal and structural reforms to promote fiscal sustainability and inclusive growth.

Kiribati: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	High
Overall risk of debt distress	High
Application of Judgment	N.A.

[1] This DSA was prepared jointly with the World Bank, and it is the first prepared under the revised Debt Sustainability Framework (DSF), approved by the Executive Board s of the IMF and the IDA in September 2017 and effective since July 2018. Debt sustainability is assessed in relation to policy-dependent debt burden thresholds.

[2] Kiribati's Composite Indicator (CI) is 2.85 based upon both the April and the October 2018 WEOs, including an update to 2017 values for the CPIA index; the "debt-carrying capacity" has been upgraded from "weak" to "medium."

BACKGROUND

1. Kiribati is a small state in the South Pacific, with limited opportunities for diversification and vulnerable to climate change. Impediments such as Kiribati's narrow production/export base (mainly limited to tuna fishing and copra) makes the country highly dependent on revenues from selling fishing licenses under various fisheries agreements and donor support. Kiribati has a sovereign wealth fund—the Revenue Equalization Reserve Fund (RERF)—which was established in 1956 from phosphate mining proceeds. Mining ceased in 1979 and, in recent years, fishing revenues have been used to replenish the fund. Sound management has seen the RERF's net capitalization value rise just above A$1 billion in August 2018—a target that was expected to be reached in 2020.

2. The country's coverage of public sector debt is the central government and social security fund. Data availability limits debt coverage: a regularly updated balance sheet for all subsectors (most notably, the SOEs) does not currently exist but recent and planned technical assistance aim to provide improvement over time. Debt guaranteed by the government is 6 percent of GDP (not included in the baseline). Public debt doubled since 2014 but has been stable around 23 percent of GDP as of end-2017. The definition of external/domestic debt is "residency-based" as local currency denominated debt (i.e., debt in Australian dollars, which is the legal tender in Kiribati) is held entirely by non-residents (see table on "Public Debt Balance as of end-2017" below).

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central government plus social security, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	6.0	Debt guaranteed by government 6 percent
4	PPP	35 percent of PPP stoc	0.0	
5	Financial market (the default value of 6 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+6) (in percent of GDP)		11.0	

1/ The default shock of 2% of GDP will be triggered for countries, whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

Kiribati: Public Debt Balance as of End-2017

Creditor	Balance
Asian Development Bank	AUD 32,225,287
International Cooperation and Development Fund, Taiwan Province of China	AUD 20,227,983

Source: Kiribati Ministry of Finance, 2018 Budget.

3. "Debt-carrying capacity" has been upgraded to "medium." Kiribati's Composite Indicator (CI) is 2.85 in both the April and the October 2018 WEOs, with an update to the CPIA index for 2017. Consequently, the requirement that two consecutive signals are needed to change the status of a country is met,[3] and these CI readings put the country in the medium debt-carrying capacity classification.[4] The relevant indicative thresholds for this category are: 40 percent for the PV of debt-to-GDP ratio, 180 percent for the PV of debt-to-exports ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt. The benchmark for the PV of total public debt for medium debt-carrying capacity is 55 percent.[5] The upgrade of the debt-carrying capacity is mainly the result of a change in methodology under the new Debt Sustainability Framework.

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 2.85	Medium 2.85	Weak 2.95

UNDERLYING ASSUMPTIONS

4. Key assumptions are consistent with the macroeconomic framework based on data provided by the authorities and estimates by staff. The 2018 real growth projection is broadly in line with projections in the last DSA. The long-term growth moderation is due to lower-than-expected infrastructure projects implementation and the worsening impact of climate change related events. Some near- to medium-term fiscal indicators have moderately improved mainly due to the upward revisions to fiscal revenues (notably further increases in fishing license fees). Regarding external finance, project support from IDA and ADF has been revised upward, reflecting the World Bank's and the ADB's planned scale-up for Pacific Island Countries. The following are the key macroeconomic assumptions used for the baseline scenario:

Kiribati: Baseline Macroeconomic Assumptions
(in percent of GDP, unless otherwise noted)

	2018 DSA: 2018-28 average	2017 DSA: 2017-27 average
Real GDP growth (change in percent)	1.9	2.1
Inflation (change in percent)	2.4	2.5
Non-interest current account deficit	-3.5	-5.4
Net FDI (negative = inflow)	1	1.2
Primary deficit	15.1	12.4

Source: IMF staff projections.

- **Real GDP and population growth** are projected to moderate over the long run. After posting a robust annual average of 5¼ percent in 2015–17, economic growth is expected to decline to around 2¼ percent in 2018 and slow down to 1¾ percent in the medium run and to slightly above 1½ percent in the long run. This reflects both the moderation of fishing revenue as the

[3] See the new "Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-income Countries," available at http://www.imf.org/en/Publications/PolicyPapers/Issues/2018/02/14/pp122617guidance-note-on-lic-dsf.

[4] It is worth noting that the CI has components that might not best reflect the characteristics of Kiribati's economy: for instance, import coverage of reserves has little relevance since Kiribati uses the Australian dollar as legal tender.

[5] The revised DSF no longer features remittance-adjusted thresholds, but rather it accounts for remittances in the CI.

current favorable conditions reverse and the potential worsening impact of climate-change-related events (further details on how the effect of climate change is incorporated in the analysis are below). Population growth is envisaged to average about 1.6 percent over the projection period (broadly in line with the United Nations' World Population Prospects).

- **Inflation** fell to 0.4 percent in 2017 (period average), primarily due to a one-off adjustment to the electricity tariff schedule but is anticipated to rise to 2.1 percent in 2018. Compared to the previous DSA, inflation is slightly lower in the medium term to take into account the lower inflation readings in the past few years, driven by passing of SOE efficiency gains to consumers. In the medium to longer terms, consumer price increases are expected to remain around 2½ percent, in line with trading partner inflation and international food and fuel price dynamics given that the bulk of Kiribati's consumer price basket comprises imported items.
- **Fiscal revenue from the fisheries** has been strong over the past five years. After lackluster performance in the late 2000s, fishing license revenue grew at an average rate of 65 percent during 2012–15. This was primarily a result of the 2012 implementation of the Vessel Day Scheme and sound management thereafter; and a stronger U.S. dollar during the same period.[6] As the effects of this boost diminished, fishing revenue declined from its 2015 peak of A$207 million and is expected to stabilize around A$135 million per year over 2019–21. Staff projections assume that fishing revenue will remain constant in real terms from 2022 onwards. This is a conservative assumption but carries a large degree of uncertainty, given unpredictability in weather conditions and migratory patterns of fish.
- The **fiscal position** improved significantly in 2013–17. In the medium to long term, however, the overall balance is expected to weaken. The deterioration would stem from lower revenue and higher expenditures. The primary balance is expected to worsen in 2019–20 due to the capex related to Air Kiribati's acquisition of two planes to expand international services. Recent double-digit growth in recurrent spending (excluding climate-change-related expenditures) is assumed to subside, with spending growing at an annual average rate of about 3 percent over the forecast horizon. The baseline scenario explicitly considers the impact of climate change on spending in line with the 2016 IMF Board Paper on "Small States' Resilience to Natural Disasters and Climate Change."[7] Climate-change-related maintenance and contingency expenditures are assumed to be less than 1 percent of GDP in 2019 and gradually reach 6 percent of GDP in 2028 and remain at that level thereafter.[8]
- **Development expenditure** is assumed to gradually decline from 55 percent of GDP in 2018 to around 20 percent of GDP in the long term as ongoing infrastructure projects are completed and the infrastructure gap is narrowed. Financing is assumed to be covered by loans from IDA and ADF (as reflected in the projected allocations by the World Bank and the ADB, and

[6] Fishing license fees are collected in U.S. dollars.

[7] Available at: https://www.imf.org/en/News/Articles/2016/12/12/PR16550-IMF-Discusses-Small-States-Resilience-to-Natural-Disasters-and-Climate-Change-and-IMF-Role.

[8] This assumption is informed by estimates from the literature. See Climate Change and Disaster Management (The World Bank, 2016), which estimates that the additional cost of coastal protection and infrastructure adaptation due to rainfall and temperature increases for Kiribati could amount to 12 percent of GDP annually by 2040. The DSA assumes that half of the costs will be borne by the budget while the rest is financed by development partners.

amounting to roughly 20 percent of development expenditure through the projection period), and by external grants until 2030 and by combinations of external grants and withdrawal of RERF resources starting in 2030.[9] The previous DSA assumed that 80 percent of development expenditure was financed by external grants up to 2022, while starting in 2023 it was financed in equal parts by external grants and external loans for the remaining period of projection.

- Kiribati is heavily reliant on **external grants**.[10] The baseline assumes that project-based grants will remain broadly stable in the near term (2018–20), in line with information from major development partners. Grants are assumed to reduce to about 15 percent of GDP (compared to a historical average of about 30 percent) in the long run.
- The **current account** surplus averaged around 30 percent of GDP in 2013–17, largely owing to the strong fishing license fees. As the growth in fisheries factor income slows down, the current account balance will narrow considerably and is expected to register an average deficit of around 1 percent of GDP in the long term.
- Kiribati's **current debt portfolio** is composed of external debt only, as all domestic debt was cleared in 2015. Therefore, the baseline and alternative scenarios do not assume any domestic debt in the short, medium, and long run.

5. The new realism tools suggest that the projections are reasonable (Figure 4). The primary balance is expected to deteriorate considerably in the next few years due to the exceptional capex noted above. However, this fiscal expansion is not expected to substantially boost growth because the spending is mainly on imports, resulting in a low multiplier. Changes to the contribution to real GDP growth from capital spending, compared to the previous DSA and as estimated by the template, again reflect the exceptional capex related to the aircraft acquisition. Public investment rates are higher while private investment rates are the same when compared to the previous DSA.

INCORPORATING NATURAL DISASTERS

6. Rising climate change challenges pose significant downside risks. Kiribati stands to lose the most from the negative effects of climate change, including but not limited to drought, higher incidences of natural disasters, loss of groundwater, and rising sea levels leading to coastal erosion. The costs of mitigating the adverse effects of climate change can partially be met by Kiribati's operating budget. Capital projects, however, require continued support from development partners.

[9] In the baseline, cash reserves are assumed to be drawn down to 5 percent of GDP before recourse to the RERF in 2030. Despite the withdrawals, the nominal balance of the RERF would remain above the A$1 billion target articulated in the authorities' fiscal strategy.

[10] In the IMF's macroeconomic framework, it is assumed that Kiribati will continue to benefit from its grant-only status. This assumption is consistent with MDBs' grant decision rules given the DSA rating and even assuming the realization of such grants and their impact on the DSA. However, in preparing the LIC-DSA, for World Bank (IDA) and other MDBs (in the case of Kiribati, the ADB), regular credit terms on all lending is assumed for all years in the projection period for which grant finance has not already been committed. This is because the DSA serves to test a country's capacity to take on IDA and ADB financing on credit terms. Hence, a clean assessment without potential grants is needed. Grants committed, on the basis of the DSA, can then be captured at the next DSA cycle.

7. The DSA baseline incorporates the costs and risks of natural disasters, again in line with the 2016 IMF Board Paper on "Small States' Resilience to Natural Disasters and Climate Change." For FY2018–23, staff's projections assume no natural disasters. This ensures that adjustment for natural disasters do not complicate near-term policy discussions. However, long-term growth projections are adjusted down by 0.1 percentage points to 1.6 percent, compared with non-disaster potential growth rate of 1.7 percent.[11] In addition, the near-term risk of a one-off extreme natural disaster is incorporated in the DSA through a tailored stress test which assumes that a one-off extreme natural disaster would decrease real GDP growth by 1.5 percent and exports by 3.5 percentage points. This aims to capture the possibility that the climate change adaptation costs (already incorporated in staff's macroeconomic framework) may turn out to be optimistic or that Kiribati becomes prone to natural disasters more than it historically has been as a result of climate change.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

8. Under the baseline scenario, Kiribati's external debt trajectory is projected to exceed the indicative threshold in the medium to long term.[12] The PV of debt-to-GDP ratio breaches the indicative benchmark (40 percent) after 2022 (Figure 1). The ratio of the PV of external-debt-to-exports ratio breaches the threshold (180 percent) at the end of the projection period. As the bulk of the projected external debt is on concessional terms, debt service will be relatively contained. However, the debt service-to-exports ratio will gradually approach the indicative threshold by the end of the projection period because of continued debt accumulation.

9. Stress tests confirm the vulnerability of debt dynamics to export market conditions as well as to macroeconomic shocks. Under the extreme test scenarios, the PV of debt-to-GDP and PV of debt-to-export ratios breach their thresholds by 2019 and 2021, respectively (Figure 1). These ratios are vulnerable to shocks emanating from exports.[13] The other stress test scenarios, including the severe natural disaster scenario, and for PV of debt -to-GDP, the contingent liabilities test, illustrate the vulnerability of the debt trajectory to external and potential domestic shocks (Table 3).

PUBLIC DEBT SUSTAINABILITY ANALYSIS

10. Public debt follows the same dynamics as external debt, given that there is no domestic debt. The tapering of the windfall fishing revenue and contingent liabilities stemming from SOEs[14]

[11] The assumed reduction is informed by the literature, see Box 3 in the 2016 Article IV Staff Report. Note that this estimate is surrounded by great uncertainty and climate change may have much larger adverse impact on growth, which the natural disaster scenario of the DSA aims to capture.

[12] The large residual in Table 1 is attributable to several factors: quality of balance of payments data, accumulation of assets in the RERF, and the partial utilization assumption regarding IDA/ADF commitments (these enter the DSA in full, but development expenditures as reflected in the overall balance are not utilizing these funds in full).

[13] For the purposes of the DSA, the exports data include fishing license fees, which would be counted as "other primary income" under conventional balance-of-payments definitions.

[14] Currently, the Government of Kiribati explicitly guarantees any obligations that are unable to be met by the Kiribati Provident Fund, as provided under Section 10 of CAP78A (Provident Fund Act 1977).

pose risks and underscore the importance of commitment to maintaining a sound fiscal position. Under the baseline scenario, the PV of total public-debt-to-GDP ratio is projected to breach its indicative benchmark (55 percent) by 2025.[15]

11. Public debt is unsustainable under the extreme shock scenario. The most extreme stress test scenario predicts the PV of public-sector-debt-to-GDP ratio breaching the threshold by 2022; and tripling in the following decade. This might be attributed to lower growth rates triggered by lower fishing revenues and higher government external borrowing for project financing.

AUTHORITIES' VIEWS

12. The authorities broadly agreed with the DSA assessment. They expressed their strong commitment to a prudent fiscal policy stance. The main elements of the authorities' fiscal strategy for 2019 are: an overall balanced budget, zero new debt, and no withdrawals from the RERF. The authorities will continue to maintain a conservative estimate of fishing revenues in the budget so that expenditure is met within the government's means, without new debt issued and without withdrawals from the RERF. The authorities affirmed that they will continue to seek grants from bilateral donors and international financial institutions with a view to keep debt levels at prudent levels. In this context, they also recognized the need to comply with the non-concessional borrowing policies for securing grant support from the World Bank and the ADB.

CONCLUSION

13. As in previous Article IV consultations, the current DSA indicates that Kiribati is at high risk of debt distress. Under the baseline scenario, the PVs of PPG external debt-to-GDP and of public debt-to-GDP would breach the indicative thresholds during most of the projection period. Kiribati faces limited scope for external borrowing. Hence, meeting the significant public spending needs and reaching the development goals depends on using the available fiscal resources in a prudent way and securing support from development partners. Vulnerabilities are exacerbated by contingent liabilities from SOEs and by climate change. Facing these challenges while maintaining fiscal sustainability requires a reinforcement of the fiscal framework and improved governance of budget outcomes and institutions. At the same time, Kiribati's vulnerability to debt distress is mitigated by a number of factors: the decline of external support will be gradual, sheltering the country from the risk of a sudden stop in foreign financing; the government has large cash buffers and continues to accumulate resources in the RERF, which could be used in case of temporary shocks; and the country currently benefits from its grant-only status.

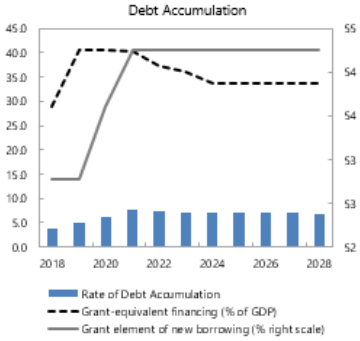
[15] The residual in the public sector DSA table reflects the volatility of fishing revenue (and RERF withdrawals / deposits). Although the fiscal position has registered a large surplus in recent years thanks to strong revenue, the surplus was mainly saved in the RERF rather than used to repay government debt.

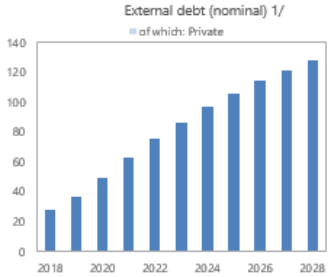
Table 1. Kiribati: External Debt Sustainability Framework, Baseline Scenario, 2015–38

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	18.5	21.6	21.0	27.9	36.9	48.7	62.9	75.3	86.5	128.0	150.4	7.6	82.2
of which: public and publicly guaranteed (PPG)	18.5	21.6	21.0	27.9	36.9	48.7	62.9	75.3	86.5	128.0	150.4	7.6	82.2
Change in external debt	3.1	3.1	-0.6	6.9	9.0	11.8	14.2	12.3	11.2	6.5	-0.6		
Identified net debt-creating flows	-44.5	-19.0	-13.3	-12.0	-6.9	-3.1	-1.7	-1.2	-0.3	6.2	24.7	-25.6	-1.1
Non-interest current account deficit	-44.4	-19.4	-13.6	-13.3	-8.5	-4.6	-3.2	-2.7	-1.6	5.4	24.5	-13.2	-2.4
Deficit in balance of goods and services	5.5	25.0	26.6	29.1	38.5	41.1	43.3	44.7	45.9	51.3	58.4	34.8	43.5
Exports	92.2	68.7	70.1	68.3	58.2	56.6	54.8	53.9	53.1	49.6	43.0		
Imports	97.7	93.7	96.7	97.4	96.6	97.6	98.1	98.6	99.0	100.9	101.4		
Net current transfers (negative = inflow)	-31.9	-25.0	-13.1	-16.5	-22.0	-18.9	-18.1	-19.1	-19.0	-18.8	-11.0	-26.4	-18.7
of which: official	-19.8	-22.3	-10.6	-14.2	-20.0	-17.1	-16.5	-17.7	-17.8	-17.1	-9.2		
Other current account flows (negative = net inflow)	-18.6	-19.4	-27.1	-25.9	-25.0	-26.8	-28.4	-28.3	-28.5	-27.0	-22.9	-21.6	-27.2
Net FDI (negative = inflow)	0.4	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.0	0.7	0.0	1.1
Endogenous debt dynamics 2/	-0.5	-0.6	-0.7	0.2	0.5	0.5	0.5	0.3	0.2	-0.2	-0.5		
Contribution from nominal interest rate	0.1	0.1	0.1	0.7	1.1	1.3	1.4	1.5	1.6	1.8	1.7		
Contribution from real GDP growth	-1.5	-0.9	-0.1	-0.5	-0.6	-0.8	-1.0	-1.2	-1.4	-1.9	-2.2		
Contribution from price and exchange rate changes	0.9	0.2	-0.7	...	...	...	...	...	...	...	...		
Residual 3/	47.7	22.1	12.7	18.9	15.9	14.9	15.9	13.6	11.6	0.3	-25.3	27.5	11.2
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	...	...	14.8	18.0	21.9	26.9	33.2	38.8	44.0	65.9	81.1		
PV of PPG external debt-to-exports ratio	...	...	21.1	26.3	37.7	47.6	60.6	71.9	82.8	132.7	188.6		
PPG debt service-to-exports ratio	0.5	0.7	0.4	1.0	2.5	2.9	3.2	3.5	3.7	4.7	10.8		
PPG debt service-to-revenue ratio	0.4	0.6	0.3	0.7	2.0	2.3	2.6	2.8	3.0	3.8	8.8		
Gross external financing need (Million of U.S. dollars)	-77.7	-33.2	-23.6	-23.1	-11.9	-3.4	0.0	1.6	4.6	28.2	146.4		
Key macroeconomic assumptions													
Real GDP growth (in percent)	10.4	5.1	0.3	2.3	2.3	2.3	2.0	1.9	1.9	1.7	1.6	2.3	2.1
GDP deflator in US dollar terms (change in percent)	-5.7	-1.0	3.6	1.5	2.6	1.9	2.3	2.5	2.6	2.9	2.9	1.0	2.2
Effective interest rate (percent) 4/	0.5	0.4	0.3	3.3	4.2	3.6	3.0	2.5	2.2	1.5	1.2	0.4	3.2
Growth of exports of G&S (US dollar terms, in percent)	8.2	-22.5	6.1	1.0	-10.5	1.4	1.0	2.9	3.0	3.0	3.1	14.5	-0.2
Growth of imports of G&S (US dollar terms, in percent)	-7.4	-0.2	7.2	4.4	4.2	5.3	4.8	5.0	5.0	4.9	4.6	5.7	4.8
Grant element of new public sector borrowing (in percent)	...	...	...	52.8	52.8	53.6	54.3	54.3	54.3	54.3	54.3	---	53.7
Government revenues (excluding grants, in percent of GDP)	106.8	84.6	96.4	93.6	73.5	71.1	69.0	66.7	65.1	60.2	52.8	66.8	73.2
Aid flows (in Million of US dollars) 5/	68.2	53.8	56.7	64.9	95.7	102.0	109.4	105.8	106.9	123.2	113.3		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	29.1	40.5	40.5	40.4	37.4	36.2	33.7	19.3	---	37.3
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	89.6	89.0	86.9	84.4	84.5	84.5	85.9	89.0	---	86.5
Nominal GDP (Million of US dollars)	178	186	193	200	210	219	229	239	250	314	487		
Nominal dollar GDP growth	4.1	4.0	3.9	3.7	5.0	4.3	4.4	4.5	4.6	4.6	4.5	3.3	4.4
Memorandum items:													
PV of external debt 7/	...	...	14.8	18.0	21.9	26.9	33.2	38.8	44.0	65.9	81.1		
In percent of exports	...	...	21.1	26.3	37.7	47.6	60.6	71.9	82.8	132.7	188.6		
Total external debt service-to-exports ratio	0.5	0.7	0.4	1.0	2.5	2.9	3.2	3.5	3.7	4.7	10.8		
PV of PPG external debt (in Million of US dollars)			28.5	36.0	46.1	59.0	75.9	92.6	109.9	206.7	395.0		
(PVt-PVt-1)/GDPt-1 (in percent)				3.9	5.1	6.1	7.7	7.3	7.2	6.8	3.5		
Non-interest current account deficit that stabilizes debt ratio	-47.6	-22.5	-13.0	-20.2	-17.5	-16.4	-17.4	-15.0	-12.8	-1.1	25.1		

Definition of external/domestic	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α = share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

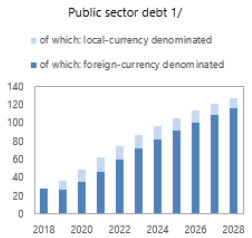
6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

Table 2. Kiribati: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–38

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**18.5**	**21.6**	**21.0**	**27.9**	**36.9**	**48.7**	**62.9**	**75.3**	**86.5**	**128.0**	**150.4**	**7.6**	**82.2**
of which: external debt	**18.5**	**21.6**	**21.0**	**27.9**	**36.9**	**48.7**	**62.9**	**75.3**	**86.5**	**128.0**	**150.4**	**7.6**	**82.2**
Change in public sector debt	3.1	3.1	-0.6	6.9	9.0	11.8	14.2	12.3	11.2	6.5	-0.6		
Identified debt-creating flows	**-48.3**	**-4.4**	**-12.2**	**19.1**	**21.6**	**20.4**	**9.3**	**6.0**	**6.5**	**9.2**	**24.1**	**-21.7**	**15.2**
Primary deficit	**-47.1**	**-3.3**	**-11.3**	**19.1**	**22.1**	**20.5**	**10.0**	**7.2**	**8.2**	**12.8**	**28.8**	**-6.4**	**13.0**
Revenue and grants	145.1	113.6	125.8	118.9	108.5	104.6	100.5	96.0	93.4	87.3	67.4	**100.5**	**96.8**
of which: grants	38.2	29.0	29.4	25.3	34.9	33.4	31.6	29.3	28.3	27.1	14.6		
Primary (noninterest) expenditure	98.0	110.3	114.5	137.9	130.6	125.1	110.5	103.1	101.5	100.1	96.3	**94.1**	**109.8**
Automatic debt dynamics	**-1.3**	**-1.1**	**-0.9**	**0.1**	**-0.5**	**-0.1**	**-0.7**	**-1.2**	**-1.7**	**-3.6**	**-4.7**		
Contribution from interest rate/growth differential	-1.6	-1.2	-0.4	-0.2	0.0	-0.2	-0.5	-0.9	-1.3	-2.5	-3.4		
of which: contribution from average real interest rate	-0.2	-0.3	-0.3	0.3	0.6	0.6	0.5	0.3	0.1	-0.6	-1.1		
of which: contribution from real GDP growth	-1.4	-0.9	-0.1	-0.5	-0.6	-0.8	-1.0	-1.2	-1.4	-2.0	-2.3		
Contribution from real exchange rate depreciation	0.3	0.1	-0.5	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**51.5**	**7.5**	**11.7**	**-11.9**	**-13.2**	**-8.5**	**4.7**	**6.1**	**4.4**	**-3.7**	**-26.0**	**23.6**	**-2.0**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	...	...	**14.8**	**18.2**	**21.9**	**27.0**	**33.2**	**38.8**	**44.0**	**65.9**	**81.1**		
PV of public debt-to-revenue and grants ratio	–	–	**11.8**	**15.3**	**20.2**	**25.8**	**33.0**	**40.4**	**47.1**	**75.5**	**120.3**		
Debt service-to-revenue and grants ratio 3/	**0.3**	**0.4**	**0.2**	**0.6**	**1.3**	**1.6**	**1.8**	**2.0**	**2.1**	**2.6**	**6.9**		
Gross financing need 4/	-46.7	-2.9	-11.0	19.7	23.6	22.2	11.8	9.0	10.1	15.1	33.5		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	10.4	5.1	0.3	2.3	2.3	2.3	2.0	1.9	1.9	1.7	1.6	**2.3**	**1.9**
Average nominal interest rate on external debt (in percent)	0.6	0.4	0.3	3.3	4.3	3.6	3.0	2.5	2.2	1.5	1.2	**0.5**	**2.5**
Average real interest rate on domestic debt (in percent)	4.9	...	...	...	...	...	...	...	...	...	...	**...**	**...**
Real exchange rate depreciation (in percent, + indicates depreciation)	2.3	0.4	-2.5	...	...	...	...	...	...	...	...	**–**	**...**
Inflation rate (GDP deflator, in percent)	13.1	0.1	0.5	1.9	3.5	1.6	2.5	2.5	2.6	2.9	2.9	**2.0**	**2.7**
Growth of real primary spending (deflated by GDP deflator, in percent)	-8.8	18.3	4.2	23.1	-3.1	-2.0	-9.8	-4.9	0.4	1.8	5.9	**8.9**	**1.0**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-50.2	-6.4	-10.7	12.1	13.2	8.7	-4.2	-5.2	-3.1	6.2	29.4	**-22.5**	**3.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

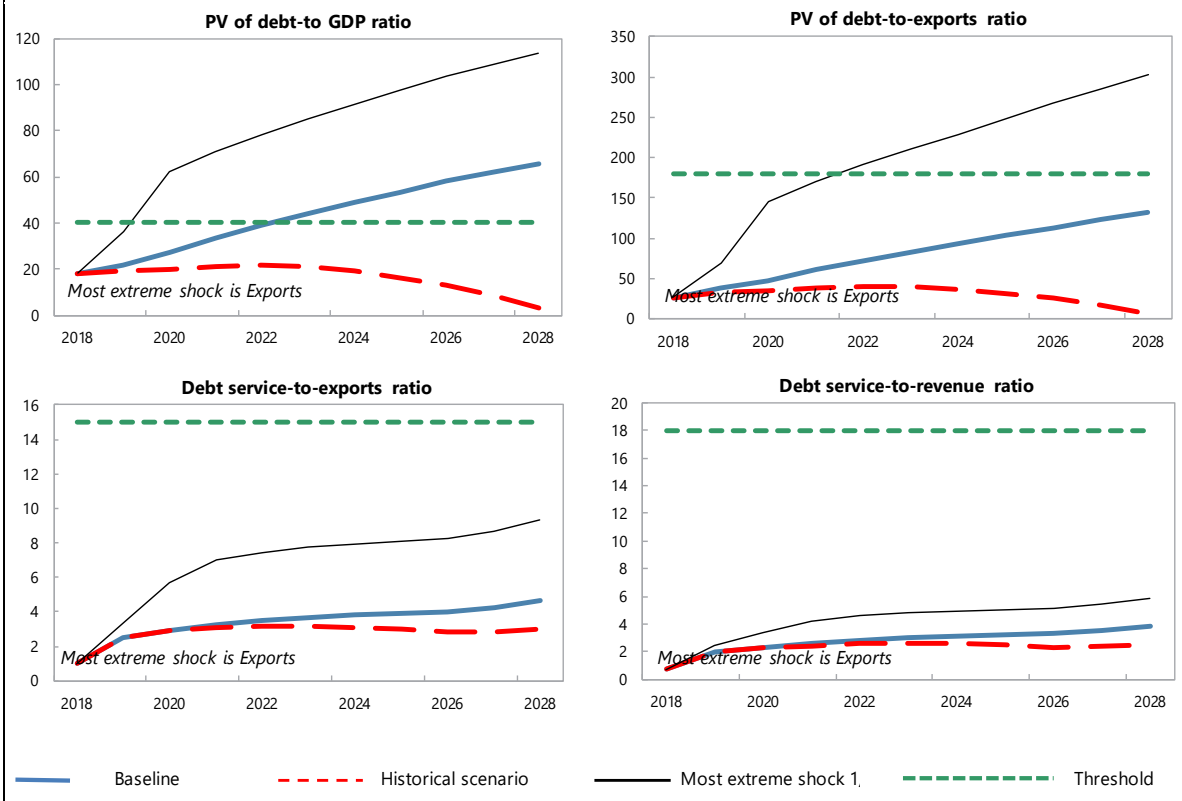
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Kiribati: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2018–28 1/



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	No	No
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.0%	1.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	36	36
Avg. grace period	9	9

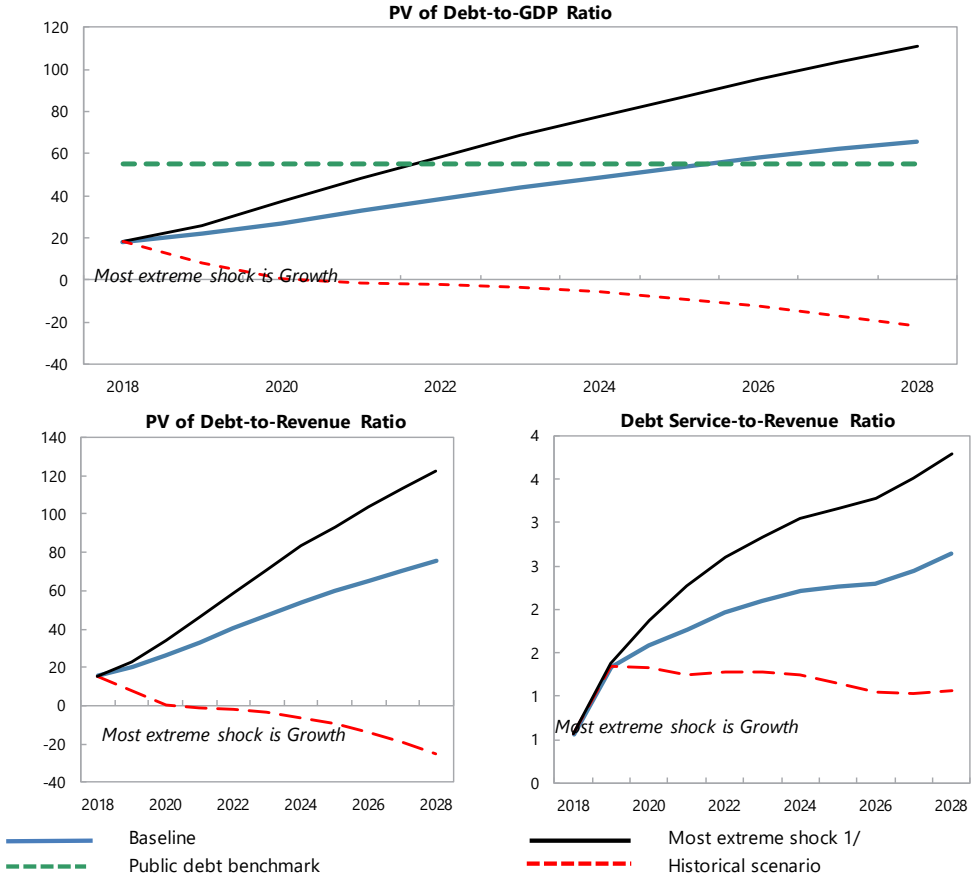
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Kiribati: Indicators of Public Debt Under Alternative Scenarios, 2018–28



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	97%	97%
Domestic medium and long-term	0%	0%
Domestic short-term	0%	3%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.0%	1.0%
Avg. maturity (incl. grace period)	36	36
Avg. grace period	9	9
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	0%	0.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Kiribati: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018–28

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	18	22	27	33	39	**44**	**49**	**54**	**58**	**62**	**66**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	18	19	20	21	21	21	19	16	13	8	3
B. Bound Tests											
B1. Real GDP growth	18	23	30	37	**44**	**50**	**55**	**60**	**65**	**70**	**74**
B2. Primary balance	18	33	**50**	**56**	**62**	**67**	**71**	**76**	**80**	**84**	**88**
B3. Exports	18	36	**62**	**71**	**78**	**85**	**91**	**97**	**103**	**109**	**114**
B4. Other flows 2/	18	29	**40**	**47**	**52**	**57**	**62**	**67**	**71**	**75**	**79**
B5. Depreciation	18	22	27	33	39	**44**	**49**	**54**	**58**	**62**	**66**
B6. Combination of B1-B5	18	36	**43**	**50**	**57**	**63**	**69**	**75**	**80**	**85**	**90**
C. Tailored Tests											
C1. Combined contingent liabilities	18	27	32	38	**44**	**49**	**54**	**59**	**63**	**67**	**71**
C2. Natural disaster	18	27	33	40	**46**	**52**	**57**	**62**	**67**	**72**	**76**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	**26**	**38**	**48**	**61**	**72**	**83**	**93**	**103**	**113**	**123**	**133**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	26	33	35	39	40	39	36	32	25	17	6
B. Bound Tests											
B1. Real GDP growth	26	38	48	61	72	83	93	103	113	123	133
B2. Primary balance	26	57	88	102	114	125	135	146	157	167	177
B3. Exports	26	69	146	171	**191**	**211**	**229**	**248**	**267**	**285**	**303**
B4. Other flows 2/	26	49	71	85	97	108	118	128	139	149	159
B5. Depreciation	26	38	48	61	72	83	93	103	113	123	133
B6. Combination of B1-B5	26	60	62	98	113	128	141	154	168	**181**	**194**
C. Tailored Tests											
C1. Combined contingent liabilities	26	47	57	70	82	93	102	113	123	133	143
C2. Natural disaster	26	48	60	75	88	100	111	123	135	146	157
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**
Debt service-to-exports ratio											
Baseline	1	2	3	3	4	4	4	4	4	4	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	1	3	3	3	3	3	3	3	3	3	3
B. Bound Tests											
B1. Real GDP growth	1	2	3	3	4	4	4	4	4	4	5
B2. Primary balance	1	2	3	4	4	4	5	5	5	5	5
B3. Exports	1	3	6	7	7	8	8	8	8	9	9
B4. Other flows 2/	1	2	3	4	4	4	4	4	4	5	5
B5. Depreciation	1	2	3	3	4	4	4	4	4	4	5
B6. Combination of B1-B5	1	3	4	5	5	5	5	5	6	6	6
C. Tailored Tests											
C1. Combined contingent liabilities	1	2	3	3	4	4	4	4	4	4	5
C2. Natural disaster	1	3	3	4	4	4	4	4	4	5	5
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	1	2	2	3	3	3	3	3	3	4	4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	1	2	2	2	3	3	3	2	2	2	2
B. Bound Tests											
B1. Real GDP growth	1	2	3	3	3	3	4	4	4	4	4
B2. Primary balance	1	2	3	3	3	4	4	4	4	4	4
B3. Exports	1	2	3	4	5	5	5	5	5	5	6
B4. Other flows 2/	1	2	3	3	3	3	3	4	4	4	4
B5. Depreciation	1	2	2	3	3	3	3	3	3	4	4
B6. Combination of B1-B5	1	2	3	3	4	4	4	4	4	5	5
C. Tailored Tests											
C1. Combined contingent liabilities	1	2	2	3	3	3	3	3	3	4	4
C2. Natural disaster	1	2	2	3	3	3	3	3	4	4	4
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

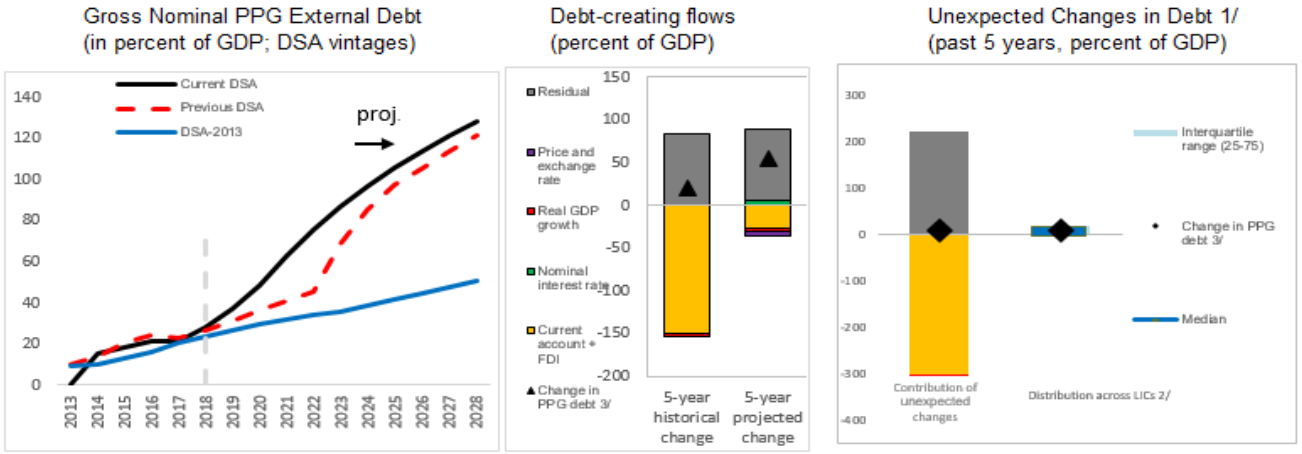
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Includes official and private transfers and FDI.

Table 4. Kiribati: Sensitivity Analysis for Key Indicators of Public Debt 2018–28

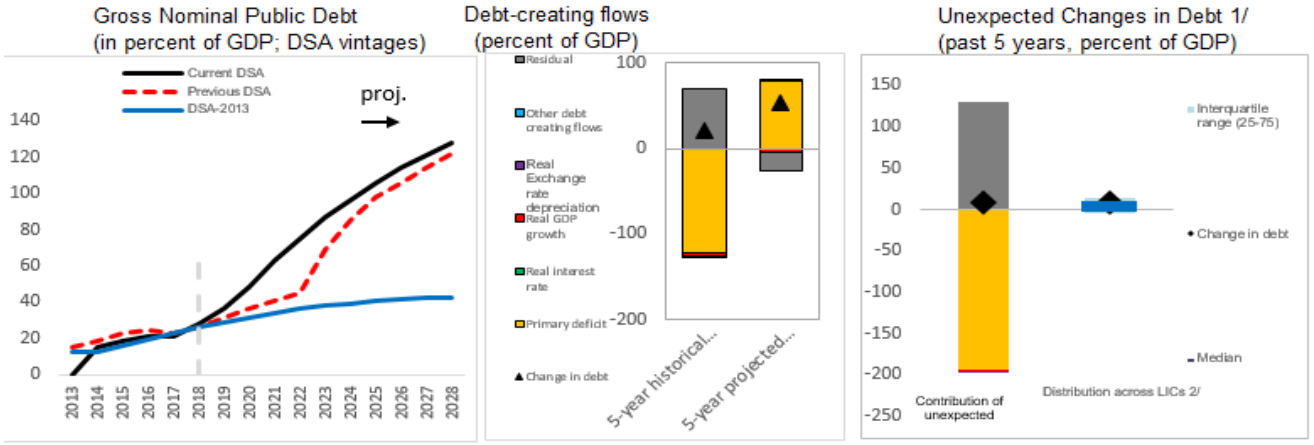
	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	18	22	27	33	39	44	49	54	**58**	**62**	**66**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	18	8	0	-1	-2	-3	-6	-9	-13	-17	-22
B. Bound Tests											
B1. Real GDP growth	18	25	37	48	**59**	**69**	**78**	**87**	**95**	**104**	**111**
B2. Primary balance	18	34	50	**56**	**62**	**67**	**71**	**76**	**80**	**84**	**88**
B3. Exports	18	29	46	52	**58**	**63**	**68**	**72**	**77**	**81**	**84**
B4. Other flows 2/	18	29	40	47	52	**57**	**62**	**67**	**71**	**75**	**79**
B5. Depreciation	18	24	24	24	25	25	26	26	27	28	28
B6. Combination of B1-B5	18	34	39	33	38	43	48	53	**57**	**61**	**65**
C. Tailored Tests											
C1. Combined contingent liabilities	18	27	32	38	44	49	54	**59**	**63**	**67**	**71**
C2. Natural disaster	18	28	33	40	46	52	**57**	**62**	**67**	**72**	**76**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	15	20	26	33	40	47	54	60	65	71	75
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	15	8	0	(1)	(2)	(4)	(6)	(10)	(14)	(19)	(25)
B. Bound Tests											
B1. Real GDP growth	15	23	34	46	59	71	83	94	104	113	123
B2. Primary balance	15	31	48	56	64	71	79	85	90	96	101
B3. Exports	15	27	44	52	60	68	75	81	86	92	97
B4. Other flows 2/	15	26	39	46	54	61	69	74	80	85	90
B5. Depreciation	15	22	23	25	27	28	29	30	31	32	33
B6. Combination of B1-B5	15	31	37	33	40	47	54	59	65	70	75
C. Tailored Tests											
C1. Combined contingent liabilities	15	25	31	38	46	53	60	66	71	76	81
C2. Natural disaster	15	25	32	40	48	55	63	69	75	81	87
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	1	1	2	2	2	2	2	2	2	2	3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	1	1	1	1	1	1	1	1	1	1	1
B. Bound Tests											
B1. Real GDP growth	1	1	2	2	3	3	3	3	3	4	4
B2. Primary balance	1	1	2	3	2	3	3	3	3	3	3
B3. Exports	1	1	2	2	2	2	3	3	3	3	3
B4. Other flows 2/	1	1	2	2	2	2	2	3	3	3	3
B5. Depreciation	1	2	2	2	2	3	3	3	3	3	3
B6. Combination of B1-B5	1	1	2	2	2	2	2	2	2	2	3
C. Tailored Tests											
C1. Combined contingent liabilities	1	1	2	2	2	2	2	2	2	3	3
C2. Natural disaster	1	1	2	2	2	2	2	2	3	3	3
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 3. Kiribati: Drivers of Debt Dynamics—Baseline Scenario



Public debt



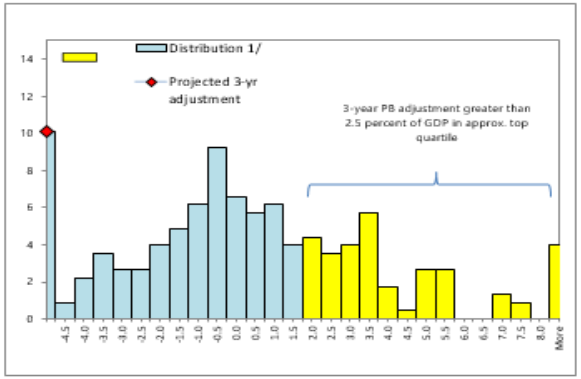
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

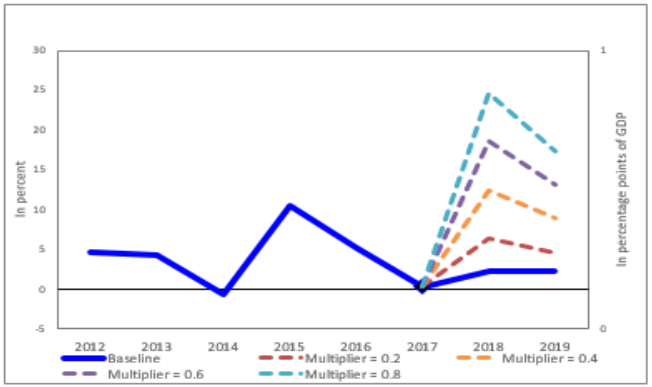
Figure 4. Kiribati: Realism Tools

3-Year Adjustment in Primary Balance(Percentage points of GDP)



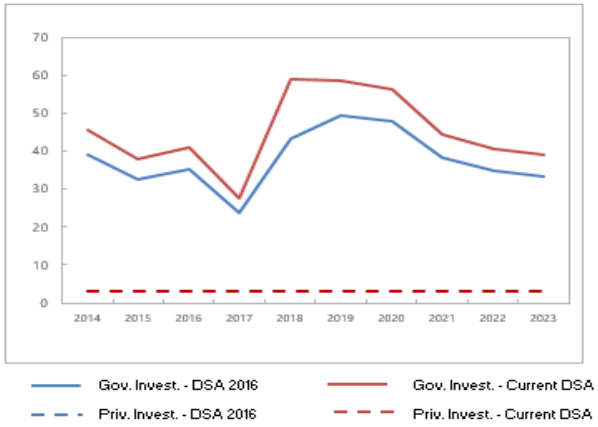
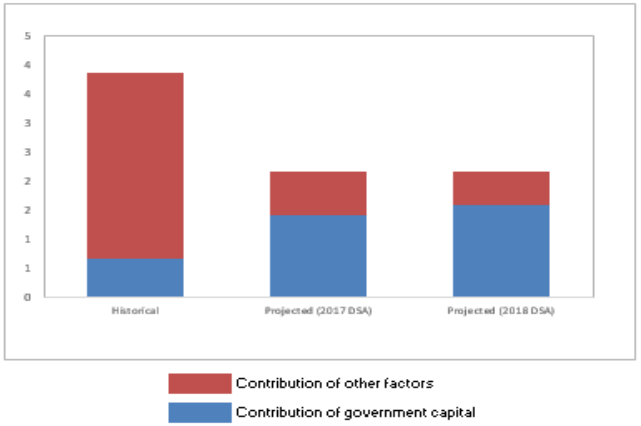
' Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent f sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)